Walid Brahim

Radar sensing and machine learning researcher · PhD candidate, Hosei University

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Summary

PhD candidate at Hosei University (Prof. Jianhua Ma, Ubiquitous Computing Lab) working on contactless health monitoring with FMCW mmWave radar, after four years as an R&D engineer in electronics manufacturing. Research interests: FMCW radar signal processing, physiological sensing, multi-sensor fusion, machine learning for time series, sensing robots, and LLM agents for research tooling. 18 IEEE/ACM papers, 6 as first author. Based in Tokyo; student visa with work permission, eligible for the Highly Skilled Professional visa on hire.

Experience

Doctoral researcher, Ubiquitous Computing Lab (Prof. Jianhua Ma)

Apr 2023 – present

Hosei University, Tokyo · defense expected Sep 2027

- Developed radar methods for respiration and heartbeat, including guided-VME inter-beat-interval estimation (IEEE CyberSciTech 2025) and seismocardiography-inspired heart monitoring (IEEE PICom 2024).
- Built the lab's platform for collecting radar data alongside reference sensors (Azure Kinect skeleton tracking, heart-rate and breathing belts), time-aligned and auto-tagged; several years of datasets.
- Measured how radar accuracy degrades with distance, posture and motion; extended the stack to eyeblink, sleep posture, in-bed state and fall detection with CNN, LSTM and Transformer models.
- Supervised about ten graduate and ten undergraduate students; collaborated with Pontosense (Canada) on data, experiments and methods.
- Teaching assistant for Python and Java; research assistant; reviewer for IEEE SWC and CyberSciTech.

Master's student, JICA ABE Initiative scholar

Sep 2020 – Sep 2022

Hosei University, Tokyo · M.Sc. in Computer and Information Sciences

- Thesis on radar-based sleep and vital-sign monitoring; its review paper and accuracy study became the base of the PhD.
- The scholarship covered the master's and a six-month internship at Automagi (AI solutions, Tokyo), 2022–2023.

Senior R&D engineer, NTIC products

Dec 2016 – present

Bomare Company, Algeria (electronics manufacturer, LG partner) · on site until 2020, continuing association since

- Responsible for the equipment, network and IT systems of two smartphone production lines.
- Designed and wrote the Stream line's production management system: raised output, unit-level traceability, fewer stoppages and errors.
- Worked directly with LG's R&D engineers, including two training periods at LG Digital Park, Korea.

Projects

MeridianOS — self-hosted research hub

TypeScript, Next.js, Node, Python, Docker, Tailscale; llama.cpp, vLLM; OpenCode, Claude Code, Codex

A private research hub I built and operate: one workspace over a fleet of devices (service host, GPU lab machine, campus H100, laptops) holding literature, notes, tasks, experiments and local LLM inference, reachable from the web, a phone, a Quest 3 headset and by voice.

- Inference gateway over local and hosted models (Qwen3-27B): client identities, serving profiles, admission control, usage ledger.
- Agent layer: written roles limiting what each agent may run, reach and change; isolated checkouts; a reviewer with hard limits.
- Tool connectors for the agents: Zotero library, task board and calendar, Telegram, voice (whisper.cpp), lab camera and microphone.
- Customised coding agents (Claude Code, Codex, OpenCode) under my own policies, hooks and MCP servers; changes are specified, built in isolation, reviewed, accepted with evidence.

Wi²Sense — radar platform for indoor health monitoring 2024 – present

Python, TI IWR6843 + DCA1000, reference sensors, HDF5, ROS 2

Records mmWave radar with reference sensors and turns the raw signal into room presence from point clouds, indoor localisation, vital signs, and sleep and activity estimates.

- Designed the three-layer stack (acquisition, engine, apps), the protocol-driven experiment mode and the offline evaluation tools.

Wi²Robot — mobile robot that positions itself to measure a person 2025 – present

ROS 2, Nav2, ROSMASTER X3 Plus (LiDAR, RGB-D), C++ and Python

Picks a viewpoint from a scene model, drives there, settles, then measures.

- Built the viewpoint-to-measurement loop on Nav2; now exploring radar-based semantic mapping of the room (Wi²Scene).

Wi²Sleep — sleep monitoring, a focused application of Wi²Sense

Single bedside 60 GHz radar; Python

Applies the Wi²Sense pipeline to one bedside radar: presence, posture, breathing and heart rate through the night, with a focus on trusted output, reliability and practicality.

Skills

Radar and signal processing	FMCW mmWave radar (TI IWR6843 + DCA1000), range-Doppler and phase-based vital-sign extraction, VMD/VME/AVMD, Azure Kinect body tracking (skeleton, joints), physiological reference sensors, synchronized recording
Machine learning	PyTorch, scikit-learn; CNN, LSTM and Transformer models for time series and radar maps; transfer learning; point clouds
Applied AI and agents	Local LLM serving (llama.cpp, vLLM), agent workflows with tool use and role policies, MCP connectors, evaluation of local models; customised coding-agent setups (Claude Code, Codex, OpenCode)
Programming	Python, TypeScript (Next.js, Node), C/C++, Java, MATLAB, SQL, LabVIEW (NI certified), VHDL
Robotics and systems	ROS 2, Nav2, LiDAR and RGB-D, Raspberry Pi, Linux, Docker, Git and CI, Tailscale, networking (Cisco CCNA)
Languages	English and French (professional), Arabic (native), Japanese (basic, studying for JLPT N4)

Education

Ph.D., Computer and Information Sciences Apr 2023 – Sep 2027 (expected)

Hosei University, Tokyo

M.Sc., Computer and Information Sciences Sep 2020 – Sep 2022

Hosei University, Tokyo · JICA ABE Initiative scholarship

M.Sc., Control and Automation 2013 – 2015

University of Blida, Algeria

B.Sc., Control and Automation 2010 – 2013

National High School for Technology (ENST), Algeria

Awards and certifications

- Best Student Paper Award, IEEE Smart World Congress 2023, for “FMCW Radar-Enabled Eyeblink Detection: An AVMD-Based Solution”
- JICA ABE Initiative scholarship, 2020–2022
- NI LabVIEW Core 1–3 (2019) · Cisco CCNA Routing and Switching (2016)

Publications (first author; 18 in total)

[1] W. Brahim, J. Ma, M. Ma, A. Qi. *FMCW Radar Efficient IBI Estimation Using Guided VME: Design and Experimental Validation*. IEEE CyberSciTech 2025.

[2] W. Brahim, J. Ma, M. Ma, Y. Luo, A. Qi, Y. Qi. *Seismocardiography-Inspired Heart Activity Monitoring Using FMCW Radar*. IEEE PICom 2024.

[3] W. Brahim, X. Zhang, J. Ma, M. Ma, A. Qi. *FMCW Radar-Enabled Eyeblink Detection: An AVMD-Based Solution*. IEEE Smart World Congress 2023, Best Student Paper Award.

[4] W. Brahim, J. Ma, M. Ma, A. Qi, Y. Luo, Y. Qi. *An Integrated Platform for Synchronous and Auto-Tagged Data Collection*. IEEE SmartWorld/UIC 2022.

[5] B. Walid, J. Ma. *Accuracy Assessment and Improvement of FMCW Radar-Based Vital Signs Monitoring Under Practical Scenarios*. IEEE DASC 2022.

[6] B. Walid, J. Ma, M. Ma, A. Qi, Y. Luo, Y. Qi. *Recent Advances in Radar-Based Sleep Monitoring: A Review*. IEEE DASC 2021.

Plus 12 co-authored papers as advisor to master’s students (2022–2025), listed on Google Scholar.